excess^3: A Pre-Specified Continuous Proxy for Order-3 Synergistic Surplus in the Systemic Tau / RECD paradigm

Methods, Null Models, and Synthetic Validation

Johel Padilla-Villanueva

Department of Environmental Health, University of Puerto Rico

ORCID: 0000-0002-5797-6931

`johelpadilla@gmail.com`

`joel.padilla2@upr.edu`

Preprint v1.0 | July 2026

Repository: github.com/johelpadilla/excess3

Abstract

High-order multivariate dependence (structure that cannot be reduced to pairwise interactions) is central to theories of integration, emergence, and critical reorganisation, yet it is difficult to estimate and easy to over-interpret. Within the Systemic Tau / Discrete Extramental Clock (RECD) paradigm, Level 3 of the nested ordinal hierarchy is intended to capture exactly this irreducible joint surplus. Here we provide a canonical, honest specification of its continuous operational form, **excess**^{3} (denoted `excess3` in code and software).

excess^3 is a hybrid of (i) synergistic excess of total correlation over a pairwise mutual-information baseline (Syn) and (ii) weighted log-ratio surprise of joint symbol tuples under a full independence model (Surp), combined with fixed a priori weights 0.6/0.4. Continuous excess^3 is the primary Level-3 readout; the binary indicator $\Phi_3 = \mathbf{1}\{\text{excess3} > \theta_3\}$ is retained only for discrete tick counts. We argue that the weights must not be optimised on the scientific dataset if falsifiability is to be preserved; that inference should use phase-shuffle surrogates on the *full* statistic (e.g. Δexcess3), not mean-versus-mean alone; that nesting of Φ_1, Φ_2, Φ_3 is operational and parallel rather than a hard logical implication; and that excess^3 measures high-order statistical dependence, not causation, and is a computable proxy rather than a full partial information decomposition (PID) atom.

We validate the construction on four synthetic arms: independent AR(1) channels (near-null), a shared-latent coupling jump (planted joint reorganisation), coupled logistic maps, and a pure pairwise control ($R = 25$ realisations, $B = 99$ phase-shuffle surrogates, $T = 1000$). Under reference hyperparameters, phase-shuffle tests on $|\Delta\text{excess3}|$ behave as predicted for null vs. planted structure: the near-null arm (G0) has mean $\Delta = +0.0028$ and median $p = 0.560$ (4% of realisations with $p < 0.05$), while planted shared-latent reorganisation (G1) yields mean $\Delta = -0.0977$, median $p = 0.010$, and 92% extreme realisations (two-sided; the sign is negative under this generator because stronger common coupling inflates pairwise baselines relative to residual multiinformation). The logistic arm (G2) is near-null in this design (mean $\Delta = -0.0081$); the pairwise control (G3) is intermediate (mean $\Delta = -0.0482$, median $p = 0.080$). At the reference binary threshold $\theta_3 = 0.08$, Φ_3 occupancy saturates on these generators because continuous excess^3 lives near $O(1)$; discrimination resides in continuous Δ. We close with estimation caveats, downstream uses, the path to excess^k, and explicit limitations. Scope is directional synthetic validation under a pre-specified protocol—not a multi-domain empirical atlas—so that later applied papers can cite a frozen Level-3 contract rather than re-derive ad hoc variants.

Keywords: excess^3, synergistic information, partial information decomposition, RECD, Systemic Tau, ordinal patterns, total correlation, surrogate testing, complex systems, multivariate dependence

1 Introduction

Complex systems often reorganise not only in the amplitude of individual observables but in the *joint* structure that binds them. Classical early-warning signals based on variance and lag-1 autocorrelation remain powerful for a class of local bifurcations [15, 4], yet they are primarily univariate and magnitude-driven. Relational and ordinal tools—Kendall concordance, Bandt–Pompe symbolic dynamics, transfer entropy, and partial information decomposition—shift attention from size of fluctuations to organisation among variables [6, 2, 16, 21].

The Systemic Tau / RECD paradigm pairs two complementary constructs [10, 9]. **Systemic Tau** (τ_s) aggregates multivariate rank concordance into a continuous thermometer of relational coherence. The **Discrete Extramental Clock** (RECD) advances an intrinsic time from hierarchical *ordinal conjunctions*: coincidence (Φ_1), persistent pairwise relations (Φ_2), and irreducible synergistic surplus (Level 3). The continuous Level-3 measure is excess^3; its thresholded form is Φ_3.

In applied reports, Level 3 is frequently the most misunderstood piece. Binary Φ_3 may remain near zero under realistic noise while continuous excess^3 still moves; weights 0.6/0.4 look arbitrary unless pre-specification is explained; surrogate tests are applied to the wrong summary; nesting is read as logical implication; and excess^3 is mistaken for a full PID synergy atom or for a causal claim. This paper is the canonical methods specification dedicated to closing those gaps.

Contributions.

1. A complete formal definition of excess^3 aligned with the reference software implementation.
2. An explicit methodological rationale for fixed a priori weights and continuous primacy over Φ_3.
3. A recommended null protocol (phase-shuffle on the full statistic).
4. A synthetic validation battery (G0–G3) with pre-specified success criteria and sensitivity checks.
5. An honest map of what excess^3 is *not*: causation, full PID, proof of strong emergence.

Roadmap. Section 2 situates excess^3 among multivariate information measures (with a comparison table). Section 3 gives the formal construction, a key-definitions box, an intuitive magnitude reading, and a worked toy window. Section 4 states boundaries. Section 5 consolidates six design decisions that are often misread in applied reports (weights, Syn vs. Surp, continuous primacy, PID scope, nulls, nesting). Section 6 details null inference. Section 7 discusses estimation practice. Section 8 reports synthetic validation. Sections 9–10 cover uses and limitations; Section 11 discusses outlook and a reporting checklist.

2 Related measures

excess^3 lives in a neighbourhood of classical and modern multivariate information tools, without claiming to replace them. Table 1 positions the proxy relative to the measures readers most often confuse with it—in particular, how it differs from PID and O-information.

Table 1: Positioning of excess^3 relative to related multivariate interdependence measures.

Measure	**Captures mainly**	**Cost**	**Relation to excess^3**
Interaction information $I(X;Y;Z)$	Synergy + redundancy (can be negative)	Low	Partially related; excess^3 targets a non-negative residual surplus rather than a signed triple interaction.
O-information / S-information	Global redundancy vs. synergy decomposition	Medium	More global; not locked to order exactly 3.
PID synergy atom	Pure synergistic atom of the PID lattice	High	Conceptual target. excess^3 is a **computable proxy** when a stable PID estimator is not feasible.
excess^3 (this work)	Hybrid order-3 synergistic surplus	Low–med.	Pre-specified continuous proxy with fixed 0.6/0.4 weights.

Brief notes. Watanabe's total correlation (multiinformation) TC $= \sum_i H(X_i) - H(X_{1:N})$ [20, 8] is the backbone of the Syn term. Averaged pairwise MI supplies the order-2 baseline that Syn subtracts. The three-variable interaction information $I(X;Y;Z)$ can be negative, a classical synergy signature [8, 19]; excess^3 aims at a related residual but is not identical to $I(X;Y;Z)$. O-/S-information separate redundancy-like and synergy-like modes globally [14]. PID lattices isolate unique, redundant, and synergistic atoms [21, 3, 5, 7]; the synergistic atom is the conceptual target of Level 3, while excess^3 remains a *pre-specified computable proxy*. Directed information flow [16] is complementary; lagged excess^3 is future work.

3 Formal definition

Key definitions for excess^3

TC (Total Correlation) Total shared information among the N variables (Watanabe–McGill multiinformation). Backbone of the Syn term.

I_{pair} Order-2 baseline: mean pairwise mutual information scaled so that it is comparable to TC.

Syn Synergistic residual: $\text{Syn} = [\text{TC} - I_{\text{pair}}]_+$. What remains after subtracting pairwise interactions.

Surp Independence surprise: how much more often joint tuples occur than expected under full independence $P(X)P(Y)P(Z)$ (product of marginals).

excess^3 Hybrid order-3 synergistic-surplus proxy: $\text{excess3} = 0.6\,\text{Syn} + 0.4\,\text{Surp}$. Weights are fixed a priori.

Φ_3 Binary tick $\mathbf{1}\{\text{excess3} > \theta_3\}$—secondary discretisation only; continuous excess^3 is primary.

3.1 Symbolic substrate

Let $X_t \in \mathbb{R}^N$ be a multivariate series. Each coordinate is encoded by Bandt–Pompe ordinal patterns of embedding dimension m and delay τ [2, 1], yielding a symbol matrix $S_t \in \{0, \ldots, m!-1\}^N$ for $t = 1, \ldots, T_{\text{eff}}$. All Level-3 quantities are estimated on a sliding window of length w ending at t, denoted $S_{[t-w+1,t]}$.

3.2 Total correlation and pairwise baseline

Write $\pi^{(i)}$ for the i-th coordinate of the windowed symbols and $H(\cdot)$ for plug-in Shannon entropy in bits. Total correlation on the window is

$$\text{TC}(t) = \sum_{i=1}^{N} H\big(\pi^{(i)}\big) - H\big(\pi^{(1)}, \ldots, \pi^{(N)}\big). \tag{1}$$

A pairwise baseline scales the mean pairwise mutual information so that it is dimensionally comparable to a multiinformation residual [10]:

$$I_{\text{pair}}(t) = (N-1) \cdot \frac{2}{N(N-1)} \sum_{1 \le i < j \le N} I\big(\pi^{(i)}; \pi^{(j)}\big) = (N-1)\,\overline{I}_{ij}(t). \tag{2}$$

3.3 Synergy residual and independence surprise

$$\text{Syn}(t) = \big[\text{TC}(t) - I_{\text{pair}}(t)\big]_+, \tag{3}$$

where $[\cdot]_+ = \max\{\cdot, 0\}$. Syn subtracts order-2 structure so that residual multiinformation is emphasised.

Independence surprise contrasts the empirical joint frequency $p_{\text{obs}}(s)$ of each joint symbol s

with the product of marginals $p_{\mathrm{ind}}(s) = \prod_{i=1}^{N} p_i(s_i)$:

$$\mathrm{Surp}(t) = \sum_s p_{\mathrm{obs}}(s) \left[\log_2 p_{\mathrm{obs}}(s) - \log_2 p_{\mathrm{ind}}(s)\right]_+. \tag{4}$$

Surp detects joint configurations that occur more often than under complete independence—tuples over-represented relative to the product of marginals. The two terms are complementary: Syn controls a pairwise MI baseline inside the multiinformation geometry (it *subtracts* order-2 structure); Surp scores co-occurrence against full factorisation. Their joint role is restated among the six design decisions of Section 5.

3.4 Hybrid excess^3 and binary Φ_3

$$\mathrm{excess3}(t) = 0.6\,\mathrm{Syn}(t) + 0.4\,\mathrm{Surp}(t). \tag{5}$$

The weights $(0.6, 0.4)$ are **heuristic constants fixed a priori** from controlled simulations *before* analysing scientific target datasets. They are not tuned by cross-validation or grid search on the dataset of interest. That choice is part of the methodological design of the paradigm: if weights are optimised on the same records used for scientific claims, excess^3 ceases to be a pre-specified test statistic and becomes a post-hoc descriptor vulnerable to overfitting.

The binary Level-3 event indicator is

$$\Phi_3(t) = \mathbf{1}\big\{\mathrm{excess3}(t) > \theta_3\big\}, \tag{6}$$

with threshold $\theta_3 > 0$ (reference values $\theta_3 \in \{0.08, 0.10\}$ in software defaults). Continuous excess^3 is the **primary** readout for magnitude, gradients, and contrasts. Φ_3 is a **secondary discretisation** for tick counts, raster-like displays, and rate analyses. Reporting only "Φ_3 never fired" without continuous curves is methodologically incomplete under noise [10].

Interpretation of magnitudes. Values of excess3(t) near zero indicate that the joint structure of the three (or N) variables is largely explainable by pairwise interactions or by independence under the plug-in window estimator. Consistently positive values away from zero (typically $\gtrsim 0.10$–0.20 depending on domain and entropy scale; *not* a universal cut-off) suggest detectable order-3 synergistic surplus. In practice, the contrast Δexcess3 (pre/post or between conditions) is usually more informative than the absolute level. Note that Δexcess3 can be *negative* even under strong joint reorganisation if increased common coupling inflates the pairwise baseline more than the multiinformational residual—as observed in the G1 synthetic arm (Section 8). Absolute scales also shift with m, w, alphabet size, and N; always interpret with a null (Section 6) rather than a floating global threshold.

3.5 Toy numerical example

To make the algebra concrete, consider a single window of length $w = 8$ with $N = 3$ already-encoded ordinal symbols (alphabet $\{0, 1\}$). The eight joint rows implement a classical XOR-like lock—each pair of coordinates is independent, but the triple is fully determined:

$\pi^{(1)}$	$\pi^{(2)}$	$\pi^{(3)}$
0	0	0
0	1	1
1	0	1
1	1	0
0	0	0
0	1	1
1	0	1
1	1	0

(each of the four distinct joint tuples appears twice: $p_{\text{obs}} = 1/4$).

Plug-in entropies (bits) give $H(\pi^{(i)}) = 1$ for each i and $H(\pi^{(1)}, \pi^{(2)}, \pi^{(3)}) = 2$, hence

$$\text{TC} = \sum_i H(\pi^{(i)}) - H_{\text{joint}} = 3 - 2 = 1.$$

Every pairwise MI vanishes ($I(\pi^{(i)}; \pi^{(j)}) = 0$), so $I_{\text{pair}} = (N-1)\overline{I}_{ij} = 0$ and

$$\text{Syn} = [\text{TC} - I_{\text{pair}}]_+ = 1.$$

For each joint tuple s, the independence product is $p_{\text{ind}}(s) = (1/2)^3 = 1/8$, while $p_{\text{obs}}(s) = 1/4$, so $\log_2(p_{\text{obs}}/p_{\text{ind}}) = 1$ and each of the four types contributes $(1/4) \cdot 1 = 0.25$ to surprise:

$$\text{Surp} = 4 \times 0.25 = 1.$$

Therefore

$$\text{excess3} = 0.6 \cdot \text{Syn} + 0.4 \cdot \text{Surp} = 0.6 \cdot 1 + 0.4 \cdot 1 = 1.0.$$

Both hybrid ingredients fire: Syn because TC exceeds the pairwise baseline, and Surp because joint co-occurrence beats full factorisation. The same arithmetic is reproduced by the companion script `scripts/toy_example.py` and by the reference `_synergy_and_surprise` routine.

Contrast: pure common driver. If instead all three coordinates are identical (only the joints $(0,0,0)$, $(1,1,1)$, and $(2,2,2)$), pairwise MI absorbs essentially all of TC under the I_{pair} scaling, so Syn $\to 0$ while Surp remains large (joint tuples are far more frequent than under independence). excess^3 then lives almost entirely on the surprise leg. This is intentional: pure redundant locking is not the same geometry as XOR-like order-3 structure, and the hybrid keeps both modes visible rather than forcing a single atom.

3.6 Role in the RECD clock

An unweighted RECD increment may combine levels as

$$\Delta\text{RECD}_0(t) = \Phi_1(t) + \Phi_2(t) + \text{excess3}(t), \tag{7}$$

or with binary Level 3, $\Phi_1 + \Phi_2 + \Phi_3$. Weighted RECD replaces coefficients by regime-dependent $\alpha_\ell(\lambda)$ modulated by a cue derived from $|\tau_s|$ [10]. The present paper does not re-litigate λ; it fixes the Level-3 building block.

Operative nesting. Conceptually one may write $\Phi_3 \supset \Phi_2 \supset \Phi_1$ to express that deeper claims are stronger. Operationally, Φ_1, Φ_2, and excess$^3/\Phi_3$ are computed **in parallel** from S_t. There is **no hard logical implication** $\Phi_1 \Rightarrow \Phi_2 \Rightarrow \Phi_3$ in the reference code. Irreducibility of order 3 is inferred *statistically* via the excess construction (Syn+Surp after removing pairwise baselines), not by forcing a chain of Boolean implications.

4 What excess^3 is not

1. **Not causation.** excess^3 quantifies high-order statistical dependence (co-occurrence beyond pairs). Causal language requires lagged or directed variants and/or interventional designs.
2. **Not a full PID lattice.** It correlates with synergistic structure in many regimes but does not replace axiomatic PID atoms [21, 3].
3. **Not a proof of strong emergence.** Alignment with weak-emergence narratives is an interpretive option under a theoretical frame; the statistic alone does not establish strong emergence.
4. **Not free of sampling constraints.** Joint symbol estimation requires adequate T relative to alphabet size and N (Section 7).
5. **Not nested by hard logic in software.** Parallel computation; statistical excess—see Section 3.6.

5 Design decisions: six fixed choices

The six points below restate methodological contracts that are easy to misread when they appear only as asides. They are deliberate design choices of the Level-3 proxy, not after-the-fact caveats.

Why 0.6/0.4. The hybrid weights $(\alpha_{\mathrm{Syn}}, \alpha_{\mathrm{Surp}}) = (0.6, 0.4)$ are **heuristic constants fixed a priori**. They encode a mild preference for residual multiinformation over independence surprise, but they are *not* estimated by cross-validation, grid search, or any other optimiser on the scientific dataset under test. Pre-specification preserves falsifiability: excess^3 remains a declared test statistic rather than a post-hoc descriptor that can be retuned until a preferred p-value appears (Section 3.4). Alternative pre-registered pairs are admissible if stated before looking at the hypothesis data; silent re-weighting is not.

Syn versus Surp. $\mathrm{Syn} = [\mathrm{TC} - I_{\mathrm{pair}}]_+$ is the non-negative residual of total correlation after subtracting an order-2 mutual-information baseline: it asks how much multiinformational organisation remains once pairwise structure has been removed. Surp is a different geometry: it scores joint symbol tuples that are more frequent than under complete independence (p_{obs} versus $\prod_i p_i$), retaining only positive log-ratio contributions. The hybrid keeps both lenses visible rather than collapsing them into a single atom (Section 3.3).

excess^3 versus Φ_3. Continuous excess3(t) is the **primary** Level-3 magnitude: trajectories, gradients, and contrasts Δexcess3 live on the continuous scale. Binary $\Phi_3 = \mathbf{1}\{\text{excess3} > \theta_3\}$ exists only for discrete tick counts, raster-like displays, and rate summaries. Occupancy of Φ_3 can saturate when θ_3 sits below the bulk of the continuous distribution; that saturation is a property of the threshold, not evidence that Level 3 is uninformative (Sections 3.4, 8.2).

Not a complete PID. excess^3 is a **computable proxy** for order-3 synergistic surplus. It is intentionally *not* a full partial information decomposition: it does not isolate unique, redundant, and synergistic atoms of a PID lattice [21, 3]. The natural theoretical extension is to replace Eq. (5) by a stable PID synergy atom when sample size and estimator quality permit; until then the hybrid remains the operational Level-3 contract (Sections 4, 11).

Null models. Inference targets the **full declared contrast**—typically $|\Delta\text{excess3}|$ under a two-sided Monte Carlo rule—not a vague comparison of means against means of means. The default surrogate is independent phase-shuffle (or any alternative that preserves relevant univariate structure while breaking cross-channel dependence). Ranking T_{obs} inside $\{T^{(b)}\}$ is the protocol; mean-versus-mean alone is an anti-pattern (Section 6).

Operative nesting. In RECD language one may speak of deeper levels as stronger claims. In software, Φ_1, Φ_2, and excess^3/Φ_3 are computed **in parallel** from the same symbolic window S_t. There is no hard Boolean implication $\Phi_1 \Rightarrow \Phi_2 \Rightarrow \Phi_3$ in the reference code. Order-3 irreducibility is inferred *statistically* through the excess construction (Syn after pairwise baseline, plus Surp), not by forcing a chain of logical gates (Section 3.6).

6 Inference: null models and Δ

6.1 Target statistics

Common Level-3 summaries include:

- trajectory excess3(t) and its windowed mean;
- contrast Δexcess3 (pre/post event or first/second half);
- optional occupancy $\mathbb{P}(\text{excess3} > \theta_3)$.

6.2 Recommended null: phase-shuffle

Independent phase-shuffle surrogates preserve each series' approximate power spectrum while destroying cross-phase relations [18]. They are appropriate when the scientific null is "no cross-variable ordinal dependence beyond what univariate spectra imply." In other words: phase-shuffle is a good default here because it keeps each channel's linear (spectral) structure while breaking the cross-channel phase relationships that feed multivariate ordinal dependence—exactly the ingredient excess^3 is designed to detect. (Any other surrogate that breaks cross-dependence while preserving the univariate features the scientific null requires is admissible if pre-specified; the non-negotiable rule is inference on the full contrast, not mean-versus-mean alone; see also Section 5.)

Protocol.

1. Compute T_{obs} on the data (e.g. Δexcess3).
2. For $b = 1, \ldots, B$, form phase-shuffle surrogates $X^{(b)}$ (independently per coordinate) and compute $T^{(b)}$.
3. Two-sided Monte Carlo p-value on absolute effect:

$$p = \frac{1 + \#\{b : |T^{(b)}| \geq |T_{\text{obs}}|\}}{B+1}.$$

Anti-pattern. Comparing only whether the observed *mean* exceeds the mean of null means, without ranking the full statistic against the null distribution of that same statistic, under-uses the surrogate ensemble and can misstate extremity under temporal dependence.

7 Estimation practice

Table 2: Reference hyperparameters used in software defaults and in Section 8. Weights are never fitted to the scientific dataset under test.

Parameter	Reference	Role
Embedding m	3	Bandt–Pompe alphabet $m!$
Delay τ	1	ordinal embedding delay
Window w	13	local joint estimation
θ_3	0.08 (cardio-like); 0.10 default	binary ticks
Weights $(\alpha_{\text{Syn}}, \alpha_{\text{Surp}})$	$(0.6, 0.4)$	**fixed a priori** from pilot simulations; never tuned on target scientific data
Stride (validation)	3	computational thinning of windows

Plug-in histograms on the finite ordinal alphabet are the reference estimator (transparent, matching open software [13]). Kernel, kNN, or regularised estimators may reduce bias for larger alphabets but change numerical scales; any change should be declared and re-validated under the same null protocol.

Bias and variance of joint estimates grow quickly with N and alphabet size and fall with effective sample size. We recommend: (i) report T, N, m, w; (ii) use surrogates for inference; (iii) when T is marginal, consider surrogate-based bias diagnostics or bootstrap intervals on Δexcess3.

8 Synthetic validation

8.1 Design

Four generative arms probe specificity and sensitivity (Table 3). All series have length $T = 1000$, $N = 3$ where applicable, event or split at $t = 500$. We use $R = 25$ realisations and $B = 99$ phase-shuffle surrogates per realisation. The full statistic is $\Delta\text{excess3} = \overline{\text{excess3}}_{\text{post}} - \overline{\text{excess3}}_{\text{pre}}$.

Table 3: Synthetic arms and directional predictions.

Arm	Generator	Prediction
G0	Independent AR(1), $N = 3$	small $\lvert\Delta\rvert$; median p not extreme
G1	Shared latent; coupling $0.05 \to 0.80$ at event	large $\lvert\Delta\rvert$; high fraction $p < 0.05$ (sign generator-dependent)
G2	Coupled logistic maps + mixed third channel after switch	weaker / design-dependent contrast
G3	Pairwise lock of channels $(0, 1)$ only; channel 2 independent	intermediate vs G0/G1 (specificity check)

Pre-specified success criteria.

- G0: median p of $\lvert\Delta\rvert$ not systematically < 0.05.
- G1: majority of realisations with $p < 0.05$ (target $\geq 80\%$).
- Continuous excess^3 should separate pre/post more stably than Φ_3 occupancy at strict θ_3.
- No claim that numerical thresholds transfer unchanged to empirical domains.

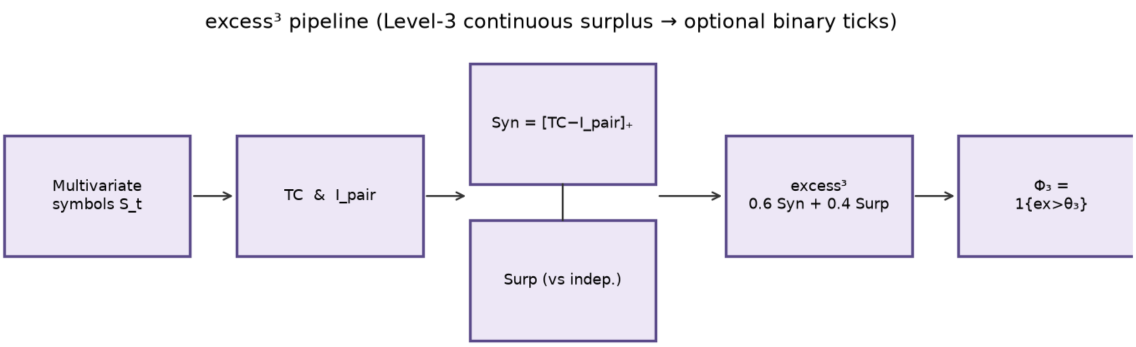

Figure 1: Map of the construction (paper overview). Multivariate series are encoded as Bandt–Pompe ordinal symbols S_t; on each window one computes total correlation (TC) and the pairwise MI baseline I_{pair}, forms the residual $\text{Syn} = [\text{TC} - I_{\text{pair}}]_+$ and the independence surprise Surp, then the hybrid continuous surplus $\text{excess3} = 0.6\,\text{Syn} + 0.4\,\text{Surp}$. Binary $\Phi_3 = \mathbf{1}\{\text{excess3} > \theta_3\}$ is an optional tick only. Inference (Section 6) applies phase-shuffle nulls to the full contrast (e.g. $\lvert\Delta\text{excess3}\rvert$); synthetic arms G0–G3 (Section 8) check directional behaviour under that contract.

8.2 Results

Table 4 summarises ensemble metrics. Figure 2 shows mean ±1 SD trajectories for G0 and G1. Figure 3 illustrates null distributions of Δ for example realisations. Figure 4 reports sensitivity to w and θ_3. Figure 5 contrasts continuous excess^3 with binary ticks on a single G1 realisation. Figure 6 summarises effect sizes and null extremity rates across arms.

Table 4: Ensemble results ($R = 25$, $B = 99$, reference hyperparameters). $\mathrm{frac}_{p<0.05}$ is the fraction of realisations with two-sided phase-shuffle $p < 0.05$ on $|\Delta\text{excess3}|$.

Arm	mean pre	mean post	mean Δ	median p	$\mathrm{frac}_{p<0.05}$	Φ_3 occ. post
G0	1.8331	1.8358	+0.0028	0.560	4%	1.000
G1	1.8308	1.7332	-0.0977	0.010	92%	1.000
G2	1.6273	1.6192	-0.0081	0.510	8%	1.000
G3	1.8374	1.7893	-0.0482	0.080	40%	1.000

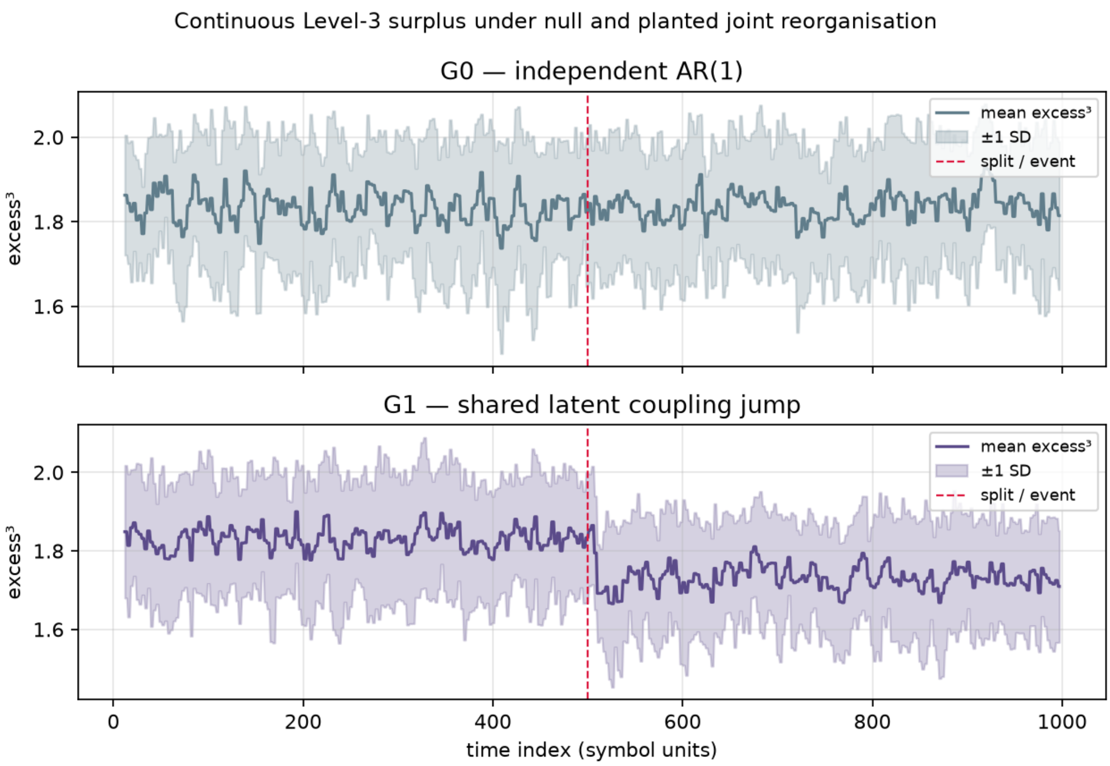

Figure 2: Mean excess$^3(t)$ ±1 SD over realisations. Vertical line: event/split. G0 remains flat; G1 shows a clear post-event shift after the shared-latent coupling jump (downward under this generator; inference uses two-sided $|\Delta|$).

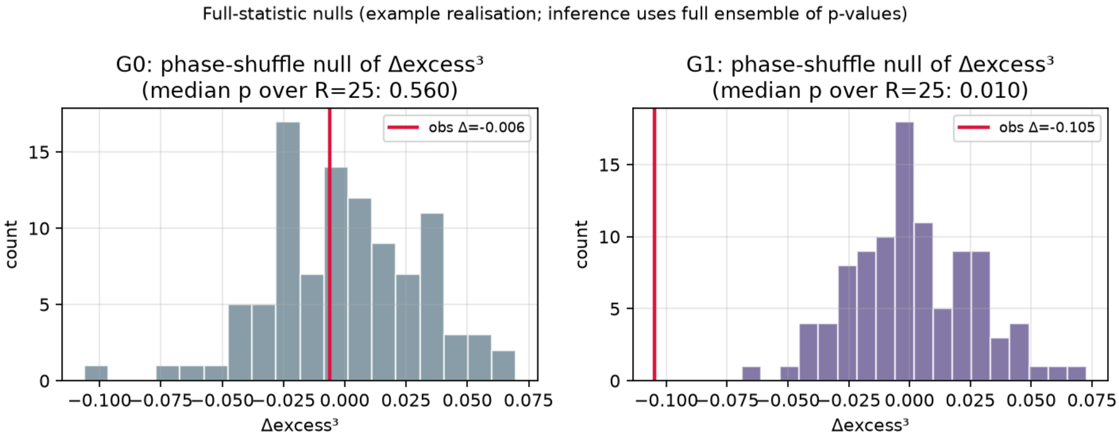

Figure 3: Phase-shuffle null histograms of Δexcess3 for an example realisation in G0 and G1, with observed Δ marked. Ensemble inference uses the full set of per-realisation p-values (Table 4).

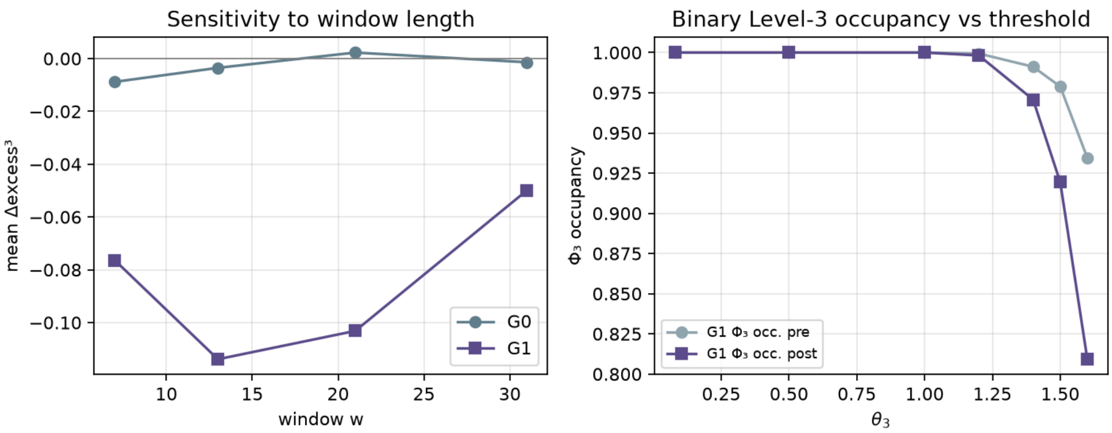

Figure 4: Left: mean Δexcess3 versus window w for G0 and G1 (qualitative separation stable across moderate w). Right: Φ_3 occupancy versus θ_3 on G1 (pre vs post). At the reference $\theta_3 = 0.08$ (vertical dashed line) occupancy saturates; only thresholds near the bulk of continuous excess^3 ($\theta_3 \gtrsim 1.2$) yield pre/post occupancy contrast—reinforcing continuous primacy under threshold uncertainty.

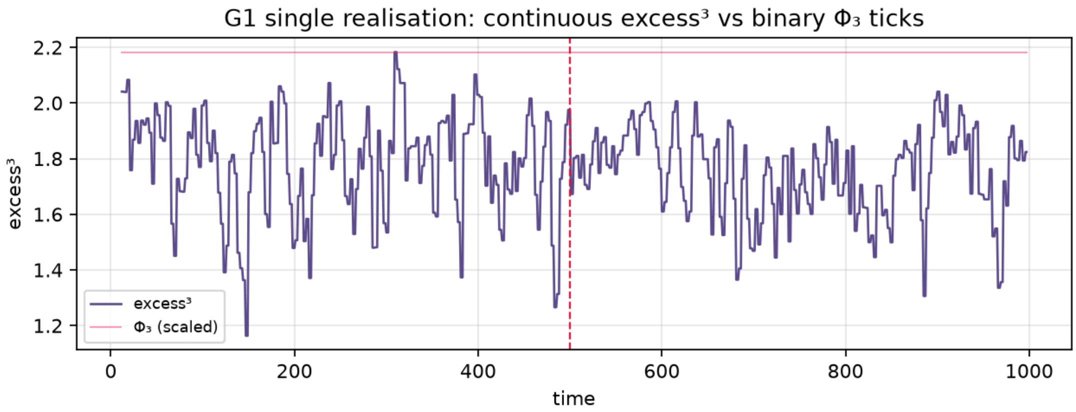

Figure 5: Single G1 realisation: continuous excess^3 versus scaled Φ_3 ticks. Binary occupancy depends on θ_3; continuous magnitude carries the primary contrast.

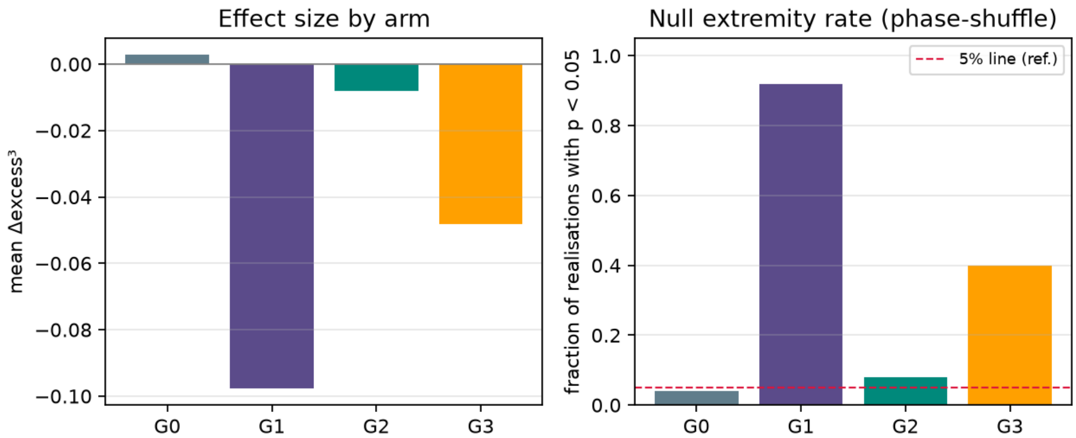

Figure 6: Left: mean Δexcess3 by arm. Right: fraction of realisations with phase-shuffle $p < 0.05$.

Interpretation. G0 behaves as a near-null: mean $\Delta = +0.0028$, median $p = 0.560$, and only 4% of realisations with $p < 0.05$ (consistent with the nominal rate under a well-calibrated null). G1 produces a clear mean shift $\Delta = -0.0977$ with median $p = 0.010$ and 92% of realisations extreme under the pre-specified two-sided phase-shuffle null, meeting the design success criterion ($\geq 80\%$ with $p < 0.05$). The *negative* sign under the shared-latent generator is methodologically informative rather than a failure: stronger common coupling after the event raises pairwise MI baselines, which can shrink the residual Syn $= [\mathrm{TC} - I_{\mathrm{pair}}]_+$ even while joint structure reorganises. Inference therefore targets $|\Delta\text{excess3}|$, not a forced positive direction. G2 (coupled logistic) is near-null in this particular design (mean $\Delta = -0.0081$, 8% extreme)—a reminder that not every nonlinear coupling inflates excess^3. G3 (pairwise only) is intermediate (mean $\Delta = -0.0482$, median $p = 0.080$, 40% extreme): weaker extremity than G1 but not pure null, consistent with partial leakage of pairwise structure into the hybrid score. Magnitudes and signs remain generator-dependent and should not be universalised.

Sensitivity panels indicate that G0/G1 separation in Δexcess3 is stable across moderate window lengths w. At the reference binary threshold $\theta_3 = 0.08$, Φ_3 occupancy is saturated on all four arms (mean continuous excess^3 near $O(1)$); raising θ_3 into the bulk produces pre/post occupancy contrast on G1, but binary tick rates then become threshold-dominated. Continuous Δexcess3 remains the primary Level-3 contrast under threshold uncertainty.

9 Downstream uses

- **High-order feature.** excess^3 or Δexcess3 as input to classifiers or regime detectors when integration/emergence proxies are of interest.
- **Regulariser.** Penalties or bonuses on order-3 synergy during model training (domain dependent; use with care).
- **Experimental contrast.** Pre/post or condition A/B comparisons with phase-shuffle (or design-based) nulls on Δexcess3.
- **Path to excess^k.** The paradigm extends to k-tuples; computational cost grows rapidly with k. excess^3 is the current expressivity–feasibility sweet spot; binary ticks become especially useful for $k > 3$ to limit dimensionality.

10 Limitations

1. Weights and the exact pairwise baseline remain heuristic, fixed for falsifiability rather than optimality.
2. Plug-in joint estimation is biased in small samples; report T and prefer surrogate inference.
3. excess^3 does not substitute for full PID when samples and estimators allow.
4. Synthetic arms are minimal; they support directional claims, not a catalogue of all bifurcations or empirical noise structures [17].
5. "Integration" language is theory-laden; the statistic measures high-order dependence, not a unique philosophical notion of emergence.

11 Discussion and outlook

This manuscript freezes the canonical Level-3 contract for the Systemic Tau / RECD paradigm: a continuous, pre-specified hybrid proxy; binary ticks as optional discretisation; phase-shuffle inference on full contrasts; parallel nesting without hard Boolean ascent; and an explicit non-causal, non-PID-complete reading. Section 5 collects the six fixed choices that most often generate applied confusion (weights, Syn/Surp, continuous primacy, PID scope, nulls, nesting). Pedagogical companions in English and Spanish may aid first contact or teaching; they are not alternative specifications.

11.1 Minimum reporting checklist

To reduce misuse and silent non-comparability across studies, any empirical report that uses excess^3 should state at least:

1. **Hyperparameters:** m, τ, w, θ_3 (if Φ_3 is shown), and stride if windows are thinned.
2. **Weights:** confirm $(0.6, 0.4)$ (or another pre-registered pair) were fixed *a priori*, not optimised on the scientific dataset.
3. **Null model:** surrogate type (default: independent phase-shuffle) and that inference was applied to the *full* contrast statistic (e.g. Δexcess3), not only mean-versus-mean.
4. **Continuous vs binary:** always report continuous excess^3 (trajectories and/or Δ); Φ_3 occupancy alone is incomplete.
5. **Sample geometry:** T (or T_{eff}), N, and alphabet size $m!$ (joint support grows as $(m!)^N$).

Natural extensions include: (i) replacing Eq. (5) by a stable PID synergy atom when estimators permit [21, 7]; (ii) lagged/directed excess^3 for asymmetric dependence; (iii) excess^k with cost-aware designs; (iv) joint reporting standards with τ_s under multi-cue RECD weighting [10]. Empirical domain papers (cardiac, ecological, neural) should cite this specification rather than re-deriving ad hoc Level-3 variants.

12 Conclusion

excess^3 is the continuous Level-3 surplus measure of the RECD hierarchy: a pre-specified combination of synergistic residual multiinformation and independence-model surprise on multivariate ordinal symbols. It is the primary magnitude for order-3 irreducibility claims; Φ_3 only counts threshold crossings. Used with honest nulls and stated limitations, it offers a practical bridge between the ideal of PID synergy and the sample sizes of real complex systems—no more, and no less.

Data and code availability

Manuscript sources, figures, ensemble JSON outputs, and validation scripts for this document family are public at github.com/johelpadilla/excess3 (folders `methods/`, `intro/`, `primer/`). A versioned archival DOI will be added upon Zenodo release. Reference Level-3 code matches the open `systemictau` / Academy Learning Tau stack [13] (Zenodo record 20576241). Reproduction of the synthetic battery needs a standard scientific Python stack (NumPy, SciPy, and related libraries) plus the Level-3 reference implementation on `PYTHONPATH`; see the repository README.

Pedagogical companions

Optional reading that does *not* redefine the statistic:

- English cross-disciplinary guide [12] (intuition, domain vignettes, teaching notes).

- Spanish dense introduction with integrated synthetic figures [11].

Cite *this* methods paper for equations, null protocol, reporting checklist, and claim-level validation.

Funding

No dedicated external grant funded this methods preprint. The broader Systemic Tau / RECD programme is developed as independent research.

Competing interests

The author declares no competing interests.

Acknowledgements

Errors of specification or exposition remain the author's.

References

[1] José M. Amigó. *Permutation Complexity in Dynamical Systems*. Springer, Berlin, Heidelberg, 2010. doi: 10.1007/978-3-642-04084-9.

[2] Christoph Bandt and Bernd Pompe. Permutation entropy: a natural complexity measure for time series. *Physical Review Letters*, 88(17):174102, 2002. doi: 10.1103/PhysRevLett.88.174102.

[3] Nils Bertschinger, Johannes Rauh, Eckehard Olbrich, Jürgen Jost, and Nihat Ay. Quantifying unique information. *Entropy*, 16(4):2161–2183, 2014. doi: 10.3390/e16042161.

[4] Vasilis Dakos, Stephen R. Carpenter, William A. Brock, Aaron M. Ellison, Vishwesha Guttal, Anthony R. Ives, Sonia Kéfi, Valerie Livina, David A. Seekell, Egbert H. van Nes, and Marten Scheffer. Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. *PLOS ONE*, 7(7):e41010, 2012. doi: 10.1371/journal.pone.0041010.

[5] Conor Finn and Joseph T. Lizier. Pointwise partial information decomposition using the specificity and ambiguity lattices. *Entropy*, 20(4):297, 2018. doi: 10.3390/e20040297.

[6] Maurice G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1/2):81–93, 1938. doi: 10.2307/2332226.

[7] Joseph T. Lizier. JIDT: An information-theoretic toolkit for studying the dynamics of complex systems. *Frontiers in Robotics and AI*, 1:11, 2014. doi: 10.3389/frobt.2014.00011.

[8] William J. McGill. Multivariate information transmission. *Psychometrika*, 19(2):97–116, 1954. doi: 10.1007/BF02289159.

[9] Johel Padilla-Villanueva. Unveiling systemic Tau: Redefining the fabric of time, stability, and emergent order across complex chaotic systems in the age of interdisciplinary discovery. Preprints.org, 2025. URL `https://doi.org/10.20944/preprints202509.1428.v1`. Foundational preprint on Systemic Tau; companion notes include `10.20944/preprints202512.0055.v1` (chaotic band).

[10] Johel Padilla-Villanueva. Systemic tau and the RECD framework: A relational theory of hierarchical ordinal conjunctions and critical transitions in complex systems, 2026. Companion framework manuscript; defines nested Φ_1–Φ_3 and sketches excess3.

[11] Johel Padilla-Villanueva. excess^3: dependencia de orden 3 y surplus sinérgico en sistemas multivariados, 2026. Dense Spanish introduction with integrated synthetic validation figures.

[12] Johel Padilla-Villanueva. Understanding excess^3: A cross-disciplinary guide to order-3 synergistic surplus, 2026. Pedagogical companion (English); does not redefine the statistic.

[13] Johel Padilla-Villanueva. systemictau: software for systemic tau and RECD analysis. Zenodo, 2026. URL `https://doi.org/10.5281/zenodo.20576241`. Open-source implementation; archival deposit.

[14] Fernando E. Rosas, Pedro A. M. Mediano, Michael Gastpar, and Henrik J. Jensen. Quantifying high-order interdependencies via multivariate extensions of the mutual information. *Physical Review E*, 100(3):032305, 2019. doi: 10.1103/PhysRevE.100.032305.

[15] Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461:53–59, 2009. doi: 10.1038/nature08227.

[16] Thomas Schreiber. Measuring information transfer. *Physical Review Letters*, 85(2):461–464, 2000. doi: 10.1103/PhysRevLett.85.461.

[17] Steven H. Strogatz. Exploring complex networks. *Nature*, 410:268–276, 2001. doi: 10.1038/35065725.

[18] James Theiler, Stephen Eubank, André Longtin, Bryan Galdrikian, and J. Doyne Farmer. Testing for nonlinearity in time series: the method of surrogate data. *Physica D*, 58(1–4): 77–94, 1992. doi: 10.1016/0167-2789(92)90102-S.

[19] Nicholas M. Timme and Christopher Lapish. A tutorial for information theory in neuroscience. *eNeuro*, 5(3):ENEURO.0052–18.2018, 2018. doi: 10.1523/ENEURO.0052-18.2018.

[20] Satosi Watanabe. Information theoretical analysis of multivariate correlation. *IBM Journal of Research and Development*, 4(1):66–82, 1960. doi: 10.1147/rd.41.0066.

[21] Paul L. Williams and Randall D. Beer. Nonnegative decomposition of multivariate information. *arXiv preprint arXiv:1004.2515*, 2010.

A Pseudocode (reference implementation)

```
# Windowed symbols S[t, i], i=1..N
TC  = sum_i H(S[:,i]) - H_joint(S)
Ipair = (N-1) * mean_{i<j} MI(S[:,i], S[:,j])
Syn = max(TC - Ipair, 0)

Surp = 0
for each joint tuple s with empirical p_obs:
    p_ind = product_i p_i(s_i)
    Surp += p_obs * max(log2(p_obs/p_ind), 0)

excess3 = 0.6*Syn + 0.4*Surp
Phi3    = 1 if excess3 > theta3 else 0
```

B Reproducibility parameters

Reference synthetic battery (from the archival script directory):

```
python3 generate_synthetic_validation.py
# optional: fill LaTeX number macros from JSON
python3 fill_numbers_from_json.py
```

Environment: Python 3 with NumPy/SciPy (and Matplotlib if regenerating figures). Ensemble defaults match the macros in `numbers.tex` / `validation_results.json` ($R = 25$, $B = 99$, $T = 1000$). Do not retune weights or the null on the same scientific dataset used for hypothesis claims (Section 5). Outputs: `figures/*.png`, `notes/validation_results.json`, `notes/validation_summary.md`. Seeds are deterministic functions of arm index and realisation index as coded in the script.